Beyond Bibliometrics: Large Language Models (LLMs) and Human Expertise Can Advance Novelty Assessment

Version 1.0, accurate as of October 30, 2026.

The dataset provided (*org_sci_sample.csv*) is an alignment of a selection of expert human scores from the Metascience Novelty Indicators Challenge survey, with the scores from the winning team – LENS (LLM-Evaluated Novelty & Significance) – for a total of 1,000 publications.

We used a stratified random sampling procedure, assigning the strata according to the categorization across the baseline distribution of novelty, specified and produced by the LENS tool. These proportions of items in each stratum are outlined in the tool's distribution curve (*Figure A1*).

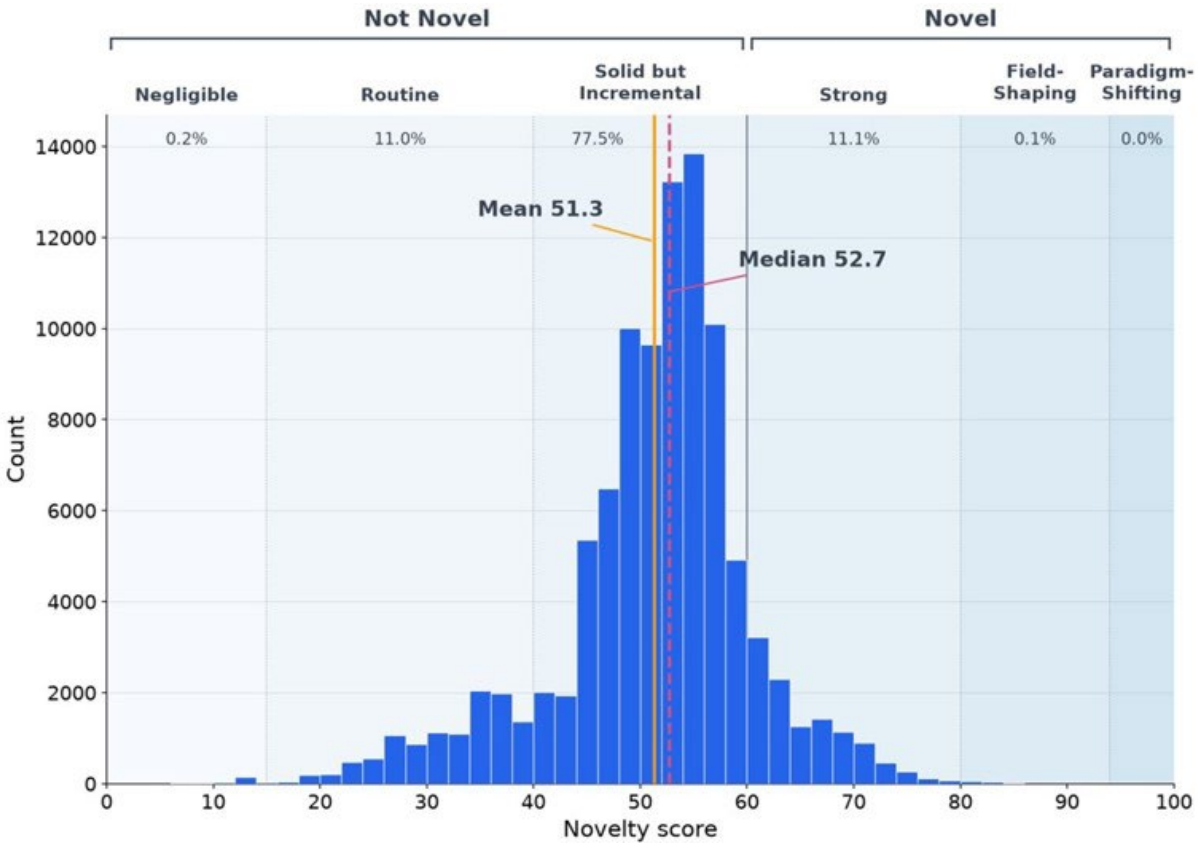


Figure A.1. – The distribution of novelty scores from the LENS novelty indicator

The publications used in this survey were sourced from OpenAlex in November 2025, with the full sampling strategy outlined in the Technical Supplement (S.14). Within the sample, all publications are also published under the CC BY 4.0 license and are publications for which LENS extracted full text. Additional metadata, outlined below in *Table A1*.

Column name	Description
Paper_ID	Project paper ID
DOI	DOI reference
OpenAlexID	OpenAlex ID
Title	The title of the publication
Date	Publication date in dd/mm/yyyy format
NumExpertRaters	Number of human raters
Expert_Median	Median of human raters' scores
Expert_Std	Standard deviation of human raters' scores
Expert_Range	Range of human raters' scores
team_score	Novelty Quant. Rating from the LENS tool
topics	Structured topic data from OpenAlex (retrieved October 2026)
Shibayama_q90	Score on Shibayama et al. (2021)
Shibayama_q100	Score on Shibayama et al. (2021)
Matsumoto	Score on Matsumoto et al. (2021)
Leahey	Score on Leahey et al. (2023)

Table A.1. – Columns in the Sample Data Set

Note. We also intend to calculate an Arts et al. (2025) score once all necessary data are available.

This is provided in the comma separated values CSV) format, as is a conversion table used to generate the sample from the full set of human-derived survey results (*Table A2*).

	Category	Cum. %	Sample size	Lower score bound	Higher score bound
Not Novel	Negligible	0-0.2%	2	1	15
	Routine	0.3-11%	110	15.1	40
	Solid but Incremental	11.1-77.5%	775	40.1	60
Novel	Strong	77.6%-99%	111	60.1	80
	Field-shaping	99-99.9%	2	80.1	94
	Paradigm-shifting	99.9%+	0	94.1	100

Table A.2. – A conversion table corresponding to Figure A1

Notes. Also provided as sampling_table.csv, without the binary novelty classification and with score boundaries transformed[1] to fit the workflow.

The process of producing this sample from the full dataset is outlined in artifact_sample.qmd (provided) and below, and supplementary files required for processing are included. Please note that only the portions of the expert human dataset used in this specific sample, and only the scores output by the winning tool for this sample, are included in this submission. Distribution plans for the full dataset are discussed in the Cover Letter; the repository then will be updated.

[1] This transformation is score x 10, which converts the score into an integer. The code for this conversion is detailed later on in this document, and was initially designed for more complex sampling strategies.

Code to generate the sample provided follows:

Code A.1 – This section comprises an entire chunk in the Quarto markdown document. It loads the required tidyverse library, sets a filepath and defines three variables used later – the winning team's internal pseudonym (a number between 1 and 30), and two numeric variables used in the sampling later.

```
rm(list=ls())

library(tidyverse)


path <- file.path("C:","Users","User","Box","Metascience","Sample data")


winning_team <- "009"

division_unit_1 = 0.1 #use to determine resolution of binning

division_unit_2 = 1000
```

Code A.2 – This is part of a chunk that loads required external data. These consist of:

1) The survey data (a sample is provided);
2) The output of the contestants' tools (in this submission, only the winning team's data is provided);
3) A table indicating how to sample according to the distribution of the tool's scores;
4) Bibliometric data from OpenAlex;
5) Additional bibliometric topic data from OpenAlex[2];
6) A spreadsheet indicating which publications had full text or only abstracts available[3]

```
setwd(path)

survey_data <- read_csv("sample_human_survey_data.csv", show_col_types = FALSE)

tool_results <- read_csv("Algorithms_Merged.csv", show_col_types = FALSE)

conversion_table <- read_csv("sampling_table.csv", show_col_types = FALSE)

OA_data <- readRDS("OA_data.Rds")

OA_subject_data <- read_csv("OA_flat_subject_data.csv", show_col_types = FALSE)

ft <- read_csv("full_text_status.csv", show_col_types = FALSE)
```

[2] Item 4 includes now-outdated topic information, prior to an OpenAlex update that improved the quality of this metadata. Item 5 re-fetches these, current as of October 2026.

[3] All items had Full Text available on the publisher website; however, some were not possible for the LENS team to bulk download due to not being accessible on either OpenAlex or Elsevier APIs, or CloudFlare blocking the team's bulk downloading pipeline.

You found this document on https://github.com/Metascience-RC/OrgSci-AI-BB-Artifact.
Please cite the data & code as [Citation to Follow].

Code A.3 – Completes the chunk started in Code A2 and:

1) Extracts the outputs of the winning team's tool;
2) Removes duplicates from the bibliometric data (at the time it was generated, OpenAlex occasionally generated multiple OpenAlex IDs for a single DOI and otherwise identical metadata);
3) Standardizes dates into dd/mm/yyyy format;
4) Creates a list of DOIs for which full text was available;

Reshapes these items for later processing

```
OA_data <- OA_data %>% select(-Keywords,-domain,-field,-subfield,-topic) %>%
  distinct(DOI, .keep_all = TRUE)

OA_data <- full_join(OA_data,OA_subject_data, by = "DOI") %>%
  select(PaperProjectID,DOI,OpenAlexID,Title,Date,topics) %>%
  rename(Paper_ID = PaperProjectID) %>%
  mutate(Date = as_date(parse_date_time(Date, orders = c("ymd","dmy"))))

ft <- ft %>% filter(str_detect(category,"fulltext_cc_by"))

ft_slim <- ft %>% select(doi)

tool_results <- tool_results %>%
  select(`DOI (as URL)`,
         contains(paste0("Converted_",winning_team)),
         contains(paste0("Raw_Certainty_",winning_team))) %>%
  rename(DOI = `DOI (as URL)`) %>%
  rename(team_score = contains("Converted")) %>%
  rename(team_bounds = contains("Certainty")) %>%
  select(-team_bounds)
```

Code A.4 – Starting a new chunk:

1) Defines output filenames;
2) Furnishes the survey data with bibliometric data and reshapes the combined dataframe;
3) Restricts the human survey data to publications that received more than one rating

```
setwd(path)

filename <- "org_sci_sample_fresh_generation.csv"
filename_survey_sample <- "org_sci_sample_survey_only.csv"

survey_data_furnished <- survey_data %>%
  select(DOI, n_raters, novel_quant_median, novel_quant_std, novel_quant_range) %>%
  rename(NumExpertRaters = n_raters, Expert_Median = novel_quant_median, Expert_StdDev = novel_quant_std,
Expert_Range = novel_quant_range)

survey_data_furnished <- survey_data_furnished %>%
  left_join(OA_data, by = "DOI") %>%
  relocate(Title,Date, .before = NumExpertRaters) %>%
  relocate(OpenAlexID, .after = DOI) %>%
  relocate(Paper_ID)

survey_data_furnished <- survey_data_furnished %>%
  filter(NumExpertRaters > 1)
```

You found this document on https://github.com/Metascience-RC/OrgSci-AI-BB-Artifact.
Please cite the data & code as [Citation to Follow].

Code A.5 – Within the same chunk:

1) Removes any tool results that correspond to publications without available full text;
2) Removes any tool results that correspond to publications with only one rater;
3) Generates a converted score (this is equivalent to score x 10 but the code was designed to allow for more complex transformations);
4) Applies category labels to the results based on the conversion table (*Table A2*);
5) Groups the data by category, and slices the number of rows specified in the conversion table.

This intermediate dataset consists of the tool's scores for the publications that will be used in the sample

```
tool_results <- tool_results %>%
  semi_join(ft_slim, by = join_by(DOI == doi)) %>%
  filter(DOI %in% survey_data_furnished$DOI)

tool_results <- tool_results %>%
    mutate(team_score_conversion = pmin(floor(team_score/division_unit_1) + 1,
      division_unit_2))

tool_results <- tool_results %>%
    mutate(category = ifelse(team_score_conversion %in%
conversion_table[[1,3]]:conversion_table[[1,4]],conversion_table[[1,1]],
                             ifelse(team_score_conversion %in% conversion_table[[2,3]]:conversion_table[[2,4]],conversion_table[[2,1]],
                                    ifelse(team_score_conversion %in%
conversion_table[[3,3]]:conversion_table[[3,4]],conversion_table[[3,1]],
                                           ifelse(team_score_conversion %in%
conversion_table[[4,3]]:conversion_table[[4,4]],conversion_table[[4,1]],
                                                  ifelse(team_score_conversion %in%
conversion_table[[5,3]]:conversion_table[[5,4]],conversion_table[[5,1]],
                                                         ifelse(team_score_conversion %in%
conversion_table[[6,3]]:conversion_table[[6,4]],conversion_table[[6,1]],"error")))))))

tool_results_sample <- tool_results %>%
    left_join(conversion_table, by = join_by(category == category_name)) %>%
    group_by(category) %>%
    group_modify(~ slice_sample(.x, n = first(.x$desired_sample))) %>%
    arrange(team_score_conversion)
```

Code A.6 – final portion of code. This removes the conversion table from the intermediate dataset, combines it with the human survey data for the relevant publications, reshapes the data for presentation, and saves both the intermediate dataset from the previous step and the final output.

```
tool_results_sample <- tool_results_sample %>%
    select(-team_score_conversion,-desired_sample,-lower_bound,-upper_bound) %>% #remove conversion table data
    relocate(DOI,team_score,category)

combined_sample <- left_join(tool_results_sample,survey_data_furnished, by = "DOI") %>%
  relocate(team_score,category, .after = Expert_Range) %>% distinct(DOI, .keep_all = TRUE) %>%
  add_column(Shibayama = "", Arts_et_al = "")

write_excel_csv(combined_sample,filename)
write_excel_csv(survey_data_furnished,filename_survey_sample)
```

Scores from pre-existing novelty metrics are calculated separately and appended to this. The code for these calculations is included in the folder novelty_indicators, and is documented separately within that folder.

REFERENCES

Arts S, Melluso N, Veugelers R (2025) Beyond citations: Measuring novel scientific ideas and their impact in publication text. *Rev. Econom. Statist.* 1-33. ePub ahead of print.

Leahey E, Lee J, Funk RJ (2023) What types of novelty are most disruptive? *American Sociological Review, 88*(3):562–597.

Matsumoto K, Shibayama S, Kang B, Igami M (2021) Introducing a novelty indicator for scientific research: Validating the knowledge-based combinatorial approach. *Scientometrics* 126(8):6891–6915.

Shibayama S, Yin D, Matsumoto K (2021) Measuring novelty in science with word embedding. *PLoS ONE 16*(7): e0254034.